SUMIT'S PORTFOLIO CODE EXPLANATION

Complete Walkthrough in Hinglish (WhatsApp Language) for Sumit Tiwari

Hey Sumit!

Ye PDF guide tere Premium Portfolio Website ke poore source code ko aasan Hinglish / WhatsApp language me samjhane ke liye banayi gayi hai. Isko read karne ke baad tujhe poore code ka flow, variables, styling rules, aur javascript animations bilkul crystal clear ho jayenge! Chalo shuru karte hain!

1. Code Ki Files Ka Overview

Pehle portfolio ka folder structure samajhte hain:

- **index.html**: Ye humare webpage ka skeleton (dhacha) hai.
- **css/style.css**: Ye humari main styling file hai jo dark theme, gradients, layout aur card shapes handle karti hai.
- **css/responsive.css**: Ye media queries contain karti hai taki screen choti hone pe navbar aur layouts scale ho sakein.
- **js/script.js**: Ye website ki base functionality (preloader, sticky nav, mobile menu, theme toggle, contact form check) chalaata hai.
- **js/animations.js**: Isme advanced visual effects (custom mouse, particle network, text typing, scroll reveals, tilt cards) likhe hue hain.

2. index.html - Skeleton Breakdown

Is html file me standard layout set kiya gaya hai, jo search engines (SEO) aur web browsers ke liye optimize hai:

Head Part: Link tags use karke FontAwesome (icons ke liye) aur Google Fonts (Poppins aur Inter fonts ke liye) import kiye gaye hain. Theme setting default dark rakhne ke liye <html lang='en' data-theme='dark'> likha hai.

Main Elements inside Body:

- **Preloader ( )**: Page open hote hi pehle ST logo scale and pulse hota hai, jab site background me load ho jati hai tab ye hide ho jata hai.
- **Particles Canvas (<canvas id='particle-canvas'>)**: Background me dynamic lines and dots draw karne ke liye use hota hai.
- **Floating Blobs ( )**: Blurred colored gradient circles jo screen par depth animations provide karte hain.
- **Custom Cursor ( )**: Mouse pointer ki jagah custom aesthetic cursor create karne ke liye tags.
- **Form validation fields**: Form inputs me placeholders ko blank (' ') rakha gaya hai taki floating labels CSS animations ke sath work kar sakein.

3. css/style.css - Design & Theme Variables

Ye file humare design system aur custom animations ko setup karti hai. Iska basic formula ye hai:

Theme Variables (:root & [data-theme='light']):

Bhai, humne variables like --bg-primary, --text-primary use kiye hain. Jab html element pe data-theme light attribute change hota hai, to in colors ki values swap ho jati hain, aur theme instant update ho jati hai.

Glassmorphic Cards Style:

Modern glassy look dene ke liye humne class use ki hai:

```
.glass-card {
  background: var(--card-bg);
  backdrop-filter: blur(16px);
  border: 1px solid var(--card-border);
}
```

Isme transparent borders aur blurred backdrop cards ko premium premium glass visual dete hain.

Hero Image glowing & floating:

Hero image me dynamic circular frame gradient and keyframe animation transition use kiya hai. Floating effect is formula pe chalta hai:

```
@keyframes floatingProfile {
  0% { transform: translateY(0px); }
  50% { transform: translateY(-15px); }
  100% { transform: translateY(0px); }
}
```

4. css/responsive.css - Mobile Responsiveness

Jab site screen 991px se choti hoti hai tab navigation responsive layout me update ho jata hai:

- **Mobile Menu**: Navbar links screen se coordinate right -100% par chale jate hain. Hamburger icon click karne pe .active class apply hoti hai, jisse panel slide hokar slide screen right: 0 pe aata hai.
- **Timeline Shifting**: Mobile screens (width <= 768px) par vertical education timeline central state se shift hokar left track side standard align ho jati hai. Timeline cards and texts 100% width occupy karte hain.

5. js/script.js - Core Interactivity

Ye file core site functionalities, scroll behavior aur form checking handle karti hai:

Preloader hide logic: Window fully load hone pe preloader element ko fade-out class dekar hide and close kiya jata hai.

Theme Switch: Theme click listener attribute read karta hai aur local storage variable change kar deta hai.

Active Link Observer (scrollActiveTracker):

Scroll check function trigger hota hai jo scroll coordinate offset read karta hai aur correct navbar indicator link pe active class toggle karta hai:

```
sections.forEach(current => {
  const sectionHeight = current.offsetHeight;
  const sectionTop = current.offsetTop - 120;
  if (scrollY > sectionTop && scrollY <= sectionTop + sectionHeight) {
    document.querySelector('.nav-menu a[href*=' + sectionId + ']').classList.add('active');
```

```
  }
});
```

Contact Form Checks: Form submission intercept ki jati hai. Validations check karte hain ki fields empty na hon aur dynamic error toggle handle ho. Submit complete hone par button loader spinner draw karta hai.

6. js/animations.js - Premium Visual Effects

Ye file website ka sabse important backend design factor hai. Isme advanced custom algorithms hain:

Custom Mouse Cursor (Double Element):

Mouse dot coordinate system clientX and clientY track karta hai. Dot instant move hota hai aur Outer Circle dynamic friction variable logic se smooth lag follow cycle generate karta hai:

```
outlineX += (mouseX - outlineX) * friction;
outlineY += (mouseY - outlineY) * friction;
```

Dynamic Canvas Particles:

Canvas particles array logic create kiya gaya hai. Har coordinate updating loop run karta hai aur collision wall boundary check rebound set hota hai. Agar do particles ke bich proximity distance 110px se kam ho, to dynamic lines draw hoti hain.

3D Mouse Tilt Card effect:

Project cards par mouse hover karne se 3D tilts generate hote hain. Mouse distance calculation coordinates card rectangle bounds delta evaluate karte hain aur standard limit angles define karte hain:

```
const normX = x / cardRect.width - 0.5;
const normY = y / cardRect.height - 0.5;
const tiltX = (normY * 12).toFixed(2);
const tiltY = -(normX * 12).toFixed(2);
card.style.transform = `perspective(1000px) rotateX(${tiltX}deg) rotateY(${tiltY}deg) scale3d(1.02,
1.02, 1.02)`;
```

Typing & Counts Scroll Reveals:

Achievements cards aur skill progress bars dynamic scrolling IntersectionObserver observe trigger state par animate aur load hote hain.

7. Summary Tip

Sumit, agar tujhe is code me koi customization karni ho (jaise project names badalna ya resume update karna), to directly HTML file me updates kar sakta hai. JavaScript and CSS automatically animations ko load aur configure kar lenge bina purane dynamic parameters ko disturb kiye!

Good luck, build premium, and make it yours! — *Antigravity code review*